

QUIZ

1. Given these declarations:

```
const char *cp = "abc";  
char UpperCase(char &ch);
```

which one of the following lines of code will compile successfully?

- a. ~~cp~~ = "xyz";
- b. cp[1] = 'b';
- c. *cp++ = '\0';
- d. char letter = UpperCase(cp[1]);

2. Output functionality needs to be provided for a class that is being designed. What is best way to reuse existing output code to provide this functionality?

- a. Make the output operator a friend of the class you are designing
- b. Provide a virtual print function for your class without using a friend function
- c. Use the C printf function within your class
- d. Use the C put function within your class

3. The linker states "unable to resolve a reference." All of the following could cause this error EXCEPT:

- a. A variable was defined as extern, but its object was not linked
- b. A procedure was called, but its object was not linked
- c. A required library was not specified
- d. A procedure is called before it is declared

4. The compiler states that a "symbol is undefined." All of the following could cause this error EXCEPT:

- a. Use of an undefined variable.
- b. Use of an undefined procedure
- c. A problem in resolving the type of an identifier
- d. Selecting debug flag when compiled

5. Given this code:

```
class A {  
public:  
    char *cpS1;  
    A::A() : cpS1(0) { }  
    A::~~A() { if (cpS1) delete cpS1; }  
};  
A a1, a2;  
a1.cpS1 = new char[64];  
strcpy(a1.cpS1, "data item one");  
a2 = a1;
```

Which one of the statements below is true?

- a. Entire class assignment with "=" is not allowed;
- b. String copy operations do not work with classes;
- c. Allocated buffer will be deleted twice;
- d. Character arrays cannot be created with operator "new".

6. All of the following statements regarding the 'new' operator are true EXCEPT:

- a. 'new' calls the appropriate constructor to initialize the memory allocated
- b. 'delete' must be explicitly called to free memory allocated by 'new'
- c. An exception is raised if memory cannot be allocated
- d. The size of the variable to be allocated need to be specified explicitly

7. A memory leak detection program is being used. Upon exit, the program indicates that memory has been left allocated. Destructors have been defined for all of the classes and the destructors are being called when it is finished with an object. What is the most likely cause of the memory leak?

- a. Compiler error
- b. A missing call to 'free' to deallocate object instances
- c. Use of base class which does not have a virtual destructor
- d. An incorrect default destructor created by the compiler

8. Given the code:

```
ostream cout;
const char *value = "123.45";
cout << "The value is " << value;
```

Which one of the following declarations for the ostream's member function "operator <<" is correct?

- a. void operator<<(const char*);
- b. ostream& operator<<(const char*);
- c. ostream& operator<<(ostream*, char*);
- d. void operator<<(ostream&, const char*);

9. All of the following are consequences of placing the keyword "const" after the declaration of a member function EXCEPT:

- a. The member function can be overloaded only with a "const" member function
- b. const must also be used in the member function definition
- c. the function cannot modify data members of the current class instance
- d. the function may not call any non-const member functions

10. Given the code:

```
template<class T> const T Abs(const T& a) {
    return a < 0 ? -a : a;
}
//...
double ch = Abs(15.3);
//...
```

- a. The code is correct as written
- b. The 'double' cannot be used as a class for templates
- c. The '<' is not defined for 'double'
- d. An 'Abs' function for 'double' must be explicitly instantiated

11. Write a C function to sort a string in **alphabet** order as generic and reusable as possible.

12. Implement C function **atoi** that converts a string to an integer.

13. 现手头上有几根绳子和一个打火机。每根绳子从一端点燃到完全燃尽都准确耗费1小时，但燃烧的速度是**不均匀**的。（也就是说，有可能烧完前半一半只耗费5分钟，而剩下另一半要耗费55分钟。）问：如何通过点燃这些绳子测量出一个75分钟的时间段？

14. 现手上**只有**一个5L的桶和一个6L的桶，如何在一个有足够水的池子里盛出3L水？

15. 小马和小杨是王老师的学生，他们知道王老师的生日M月D日是以下日期之一，但不知道具体是哪一天。王老师把M告诉了小马，把D告诉了小杨，然后问他们是否知道他的生日是哪天。小马说“如果我不知道，小杨也不知道”，小杨说“我刚才还不知道，但你说完这句话我就知道了”，小马接着说“那我现在也知道了”。问：王老师的生日是哪天：

- a. 3, 4
- b. 3, 5
- c. 3, 8

- d. 6, 4
- e. 6, 7
- f. 9, 1
- g. 9, 5
- h. 12, 1
- i. 12, 2
- j. 12, 8

16. 有500瓶酒，其中只有一瓶有毒。现在用小白鼠进行实验，小白鼠只要服用任意量有毒酒就会在24小时内死亡。问最少要用多少只小白鼠进行实验才能在24小时内检测出哪瓶药水有毒？

17. 小明一家人在黑夜过桥，需要点灯且只有一盏灯。小明过桥需要1秒，爸爸需要3秒，妈妈需要6秒，奶奶需要8秒，爷爷需要12秒，桥一次只能通过2人，而灯只能点燃30秒。如何让全家人安全过桥？

18. 某人去玩具店买小熊，小熊进价为20，卖价是30，付给老板100元，老板没有零钱，就在卖鞋的人那里换了100元零钱，找给卖家70，卖鞋的发现那100元是假币，于是找老板换回了100元，请问玩具店老板损失了多少钱？

19. 三个公司员工出差住店，每人每天是150元，共450元，店老板优惠返了50元让小工退给三个职员，小工从中扣了20元，将剩下的30元返给了三个员工，每人返回10元，这样三个人相当于每人只拿了140元，三个人加起来是420元，加上小工拿走的20元，一共是440元，请问：另外的10元哪去了？

20. 使用类似链表的方式实现一个模板队列（`template <typename T> class queue`），不要使用STL等现成的容器。